

Budapest University of Technology and Economics
Faculty of Transportation Engineering and Vehicle Engineering
Department of Transport Technology and Economics

Intelligent Transportation Systems

2025/2026 Spring

2. Assignment

GIS Assignment

Submitted by:

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1. Executive Summary

This report documents the end-to-end GIS workflow used to evaluate the topographic suitability of Bursa's urban road network for bicycle transportation. By integrating OpenStreetMap (OSM) vector data with Digital Elevation Models (DEM), we calculated precise longitudinal slopes for every road segment. This data-driven approach identifies critical corridors for sustainable transport planning and highlights areas where terrain complexity necessitates specialized infrastructure interventions.

2. Data Acquisition & Processing

2.1. Vector Data Extraction (QuickOSM)

The primary road network was acquired via the **QuickOSM** plugin. The query was focused on the highway key within the Bursa administrative boundary. To maintain data integrity, specific attributes such as fid (Feature ID), name, and highway typewere preserved for the final relational join. The output map can be seen on *Figure 1*.

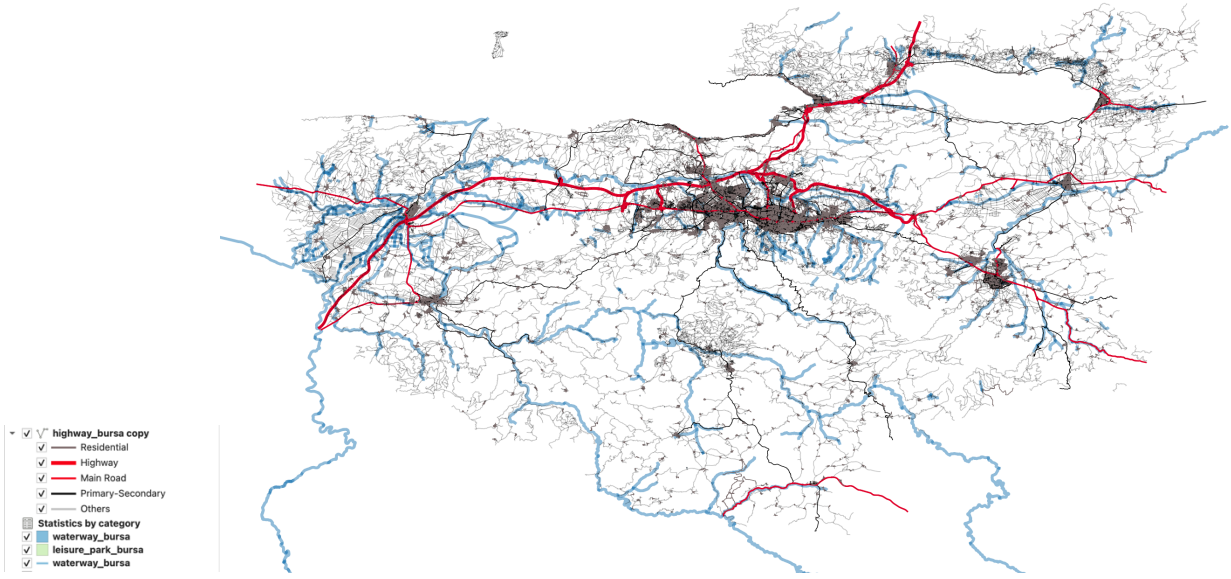


Figure 1: QuickOSM Output Map

2.2. Raster Data Acquisition (DEM)

A **Digital Elevation Model (DEM)** was utilized as the primary topographic reference. This raster data provides the necessary vertical (Z) coordinates required to calculate elevation changes across the 2D planimetric road network. SRTM Downloader extension is used as can be seen on *Figure 2*, and the DEM output map can be seen on *Figure 3*.

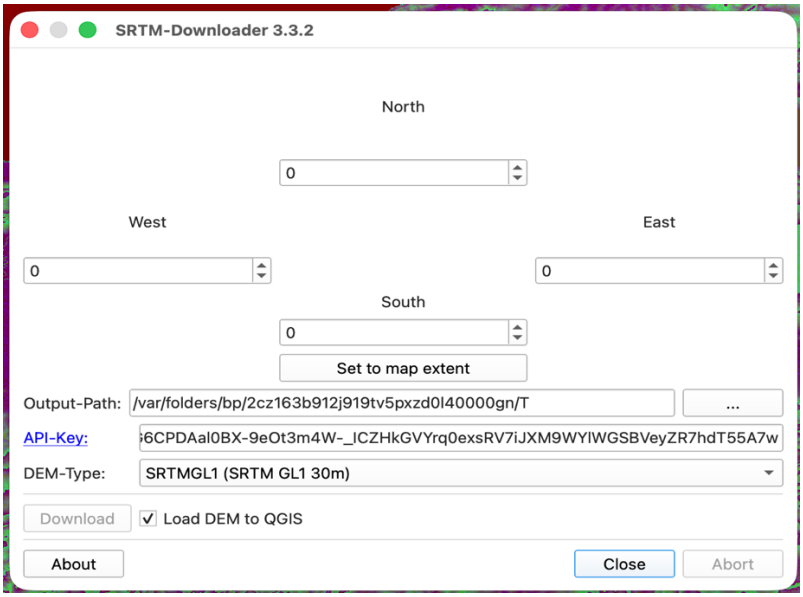


Figure 2: SRTM-Downloader Extension UI

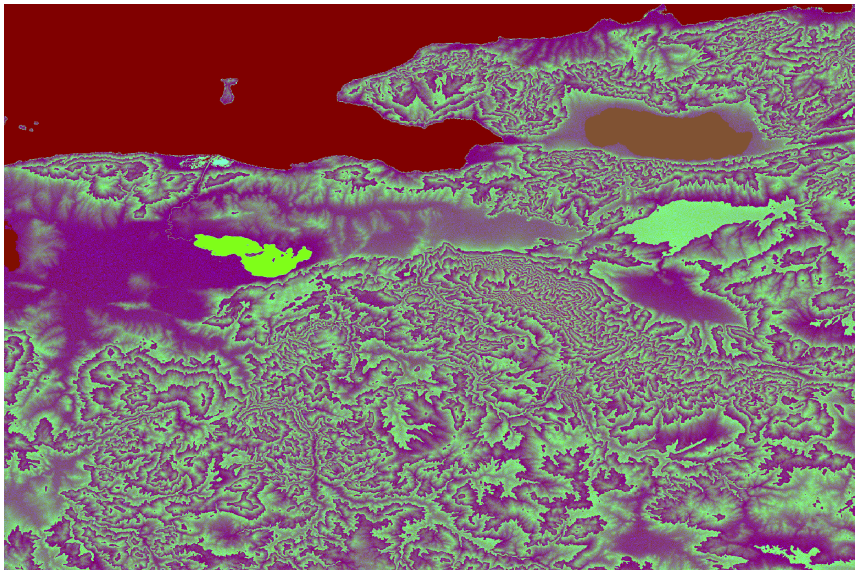


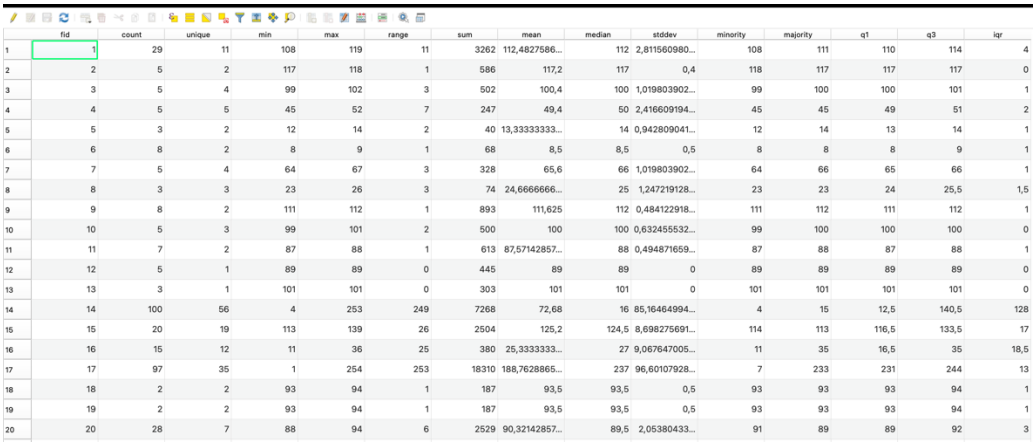
Figure 3: DEM Output

3. Technical Methodology

Phase I: Geometric Discretization Road segments were discretized into individual vertices using the Extract Vertices tool. This step is critical because elevation changes vary within a single road segment; by converting lines to points, we can sample the exact height at every node.

Phase II: Point Sampling (Z-Value Assignment) Using the Sample Raster Values tool, the DEM height information was transferred to the extracted vertices. Each point was assigned an attribute reflecting its altitude in meters.

Phase III: Statistical Aggregation To return the data to a "segment-based" format, the Statistics by Categories tool was employed. Points were grouped by their unique **Feature ID (FID)** to determine the **Minimum (min_z)** and **Maximum (max_z)** elevation for each road segment. The output statiscics output can be seen on *Figure 4*.



	fid	count	unique	min	max	range	sum	mean	median	stddev	minority	majority	q1	q3	iqr
1	1	29	11	108	119	11	3262	112,4827586...	112	2,811560980...	108	111	110	114	4
2	2	5	2	117	118	1	586	117,2	117	0,4	118	117	117	117	0
3	3	5	4	99	102	3	502	100,4	100	1,019803902...	99	100	100	101	1
4	4	5	5	45	52	7	247	49,4	50	2,416609194...	45	45	49	51	2
5	5	3	2	12	14	2	40	13,33333333...	14	0,942809041...	12	14	13	14	1
6	6	8	2	8	9	1	68	8,5	8,5	0,5	8	8	8	9	1
7	7	5	4	64	67	3	328	65,6	66	1,019803902...	64	66	65	66	1
8	8	3	3	23	26	3	74	24,6666666...	25	1,247219128...	23	23	24	25,5	1,5
9	9	8	2	111	112	1	893	111,625	112	0,484122918...	111	112	111	112	1
10	10	5	3	99	101	2	500	100	100	0,632455532...	99	100	100	100	0
11	11	7	2	87	88	1	613	87,57142857...	88	0,494871659...	87	88	87	88	1
12	12	5	1	89	89	0	445	89	89	0	89	89	89	89	0
13	13	3	1	101	101	0	303	101	101	0	101	101	101	101	0
14	14	100	56	4	253	249	7268	72,68	16	85,16464994...	4	15	12,5	140,5	128
15	15	20	19	113	139	26	2504	125,2	124,5	8,698275691...	114	113	116,5	133,5	17
16	16	15	12	11	36	25	380	25,3333333...	27	9,067647005...	11	35	16,5	35	18,5
17	17	97	35	1	254	253	18310	188,7628865...	237	96,60107928...	7	233	231	244	13
18	18	2	2	93	94	1	187	93,5	93,5	0,5	93	93	93	94	1
19	19	2	2	93	94	1	187	93,5	93,5	0,5	93	93	93	94	1
20	20	28	7	88	94	6	2529	90,32142857...	89,5	2,05380433...	91	89	89	92	3

Figure 4: Statistics Output File UI

4. Engineering Calculations & Expression Logic

The slope percentage was derived using the **Field Calculator**. The formula accounts for the vertical rise over the horizontal run:

Slope (%) = [(max_z - min_z) / Length] × 100

The $length function in QGIS provided the planimetric distance. The resulting values were then used to classify the network into cycling-friendly and challenging zones. The calculation can be seen on *Figure 5*.

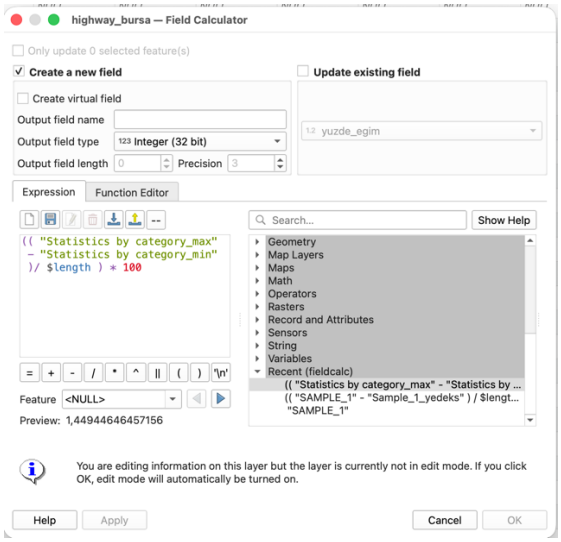


Figure 5: Field Calculator Area

5. Challenges Encountered & Technical Solutions

- **Problem 1: Function Incompatibility (Parser Errors)**
  - *Description:* Initial attempts to use built-in geometric functions like max_z($geometry) failed due to local QGIS parser errors.
  - *Solution:* Developed a three-stage workaround involving "Extract Vertices," "Sample Raster Values," and "Statistics by Categories" to manually derive vertical differences.
- **Problem 2: Computational Performance (Processing Lag)**
  - *Description:* The high density of vertices caused the processing to stall at 50%.
  - *Solution:* Optimized the dataset by using **"Points along geometry"** with a 100-meter fixed interval, reducing the data volume by 80% while maintaining statistical accuracy.
- **Problem 3: Data Noise (Highways)**
  - *Description:* Initial maps included highways (motorways) which are flat but inaccessible to cyclists.

- *Solution:* Applied a Rule-based Filter to exclude motorway, trunk, and primary roads from the final visualization.

6. Classification Standards

Classification rules are decided as in *Table 1*, the rule setting can be seen on *Figure 6*.

Table 1: Classification Rules

Classification	Slope Range	User Experience	Color Coding
Level A (Ideal)	0% - 2%	Effortless; suitable for all ages.	Green
Level B (Moderate)	2% - 5%	Requires average physical fitness.	Yellow
Level C (Challenging)	> 5%	Steep; suitable for e-bikes or sport.	Red

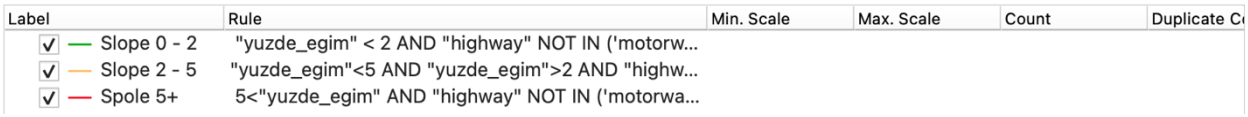


Figure 6: Classification Rules in QGIS UI

7. Conclusion

As can be seen on *Figure 7*, in the Nilüfer district (western side of the city), the analysis confirms a high percentage of "Level A" roads, making it the primary candidate for an integrated bike-sharing system. Conversely, the slopes leading to the Uludağ foothills (southern side of the city) consistently showed >8% gradients, justifying the need for electric-assist vehicle infrastructure in those corridors.

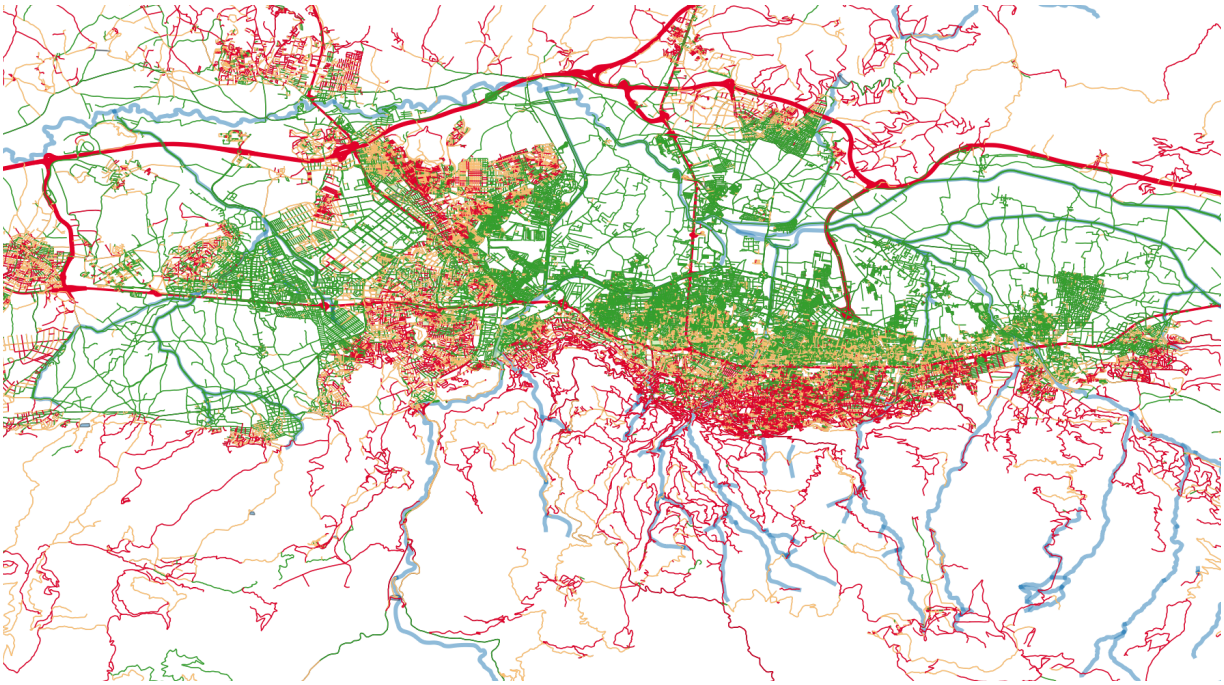


Figure 7: City Center Biking Evaluation

Final evaluation of the city can be seen on *Figure 8* and *Figure 9*. In the first figure Trunk, Trunk_link, Highway and Highway_link roads are deleted. On the second figure those deleted roads are added with thick red symbol.

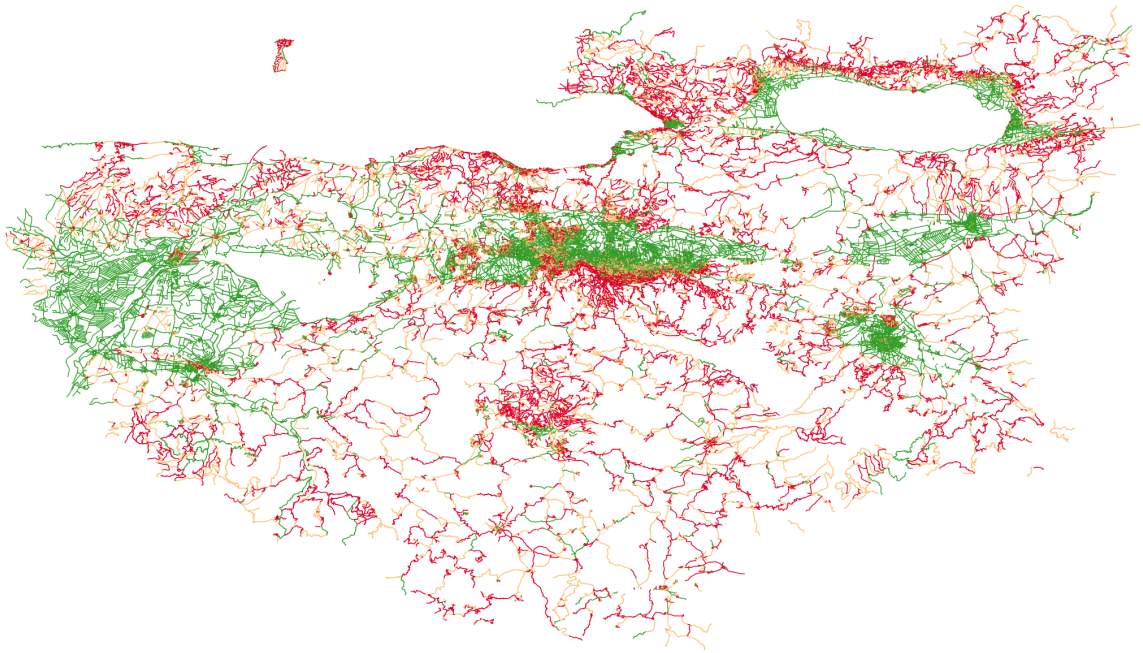


Figure 8: Bursa Whole City Biking Evaluation (Without Highways and Trunks)

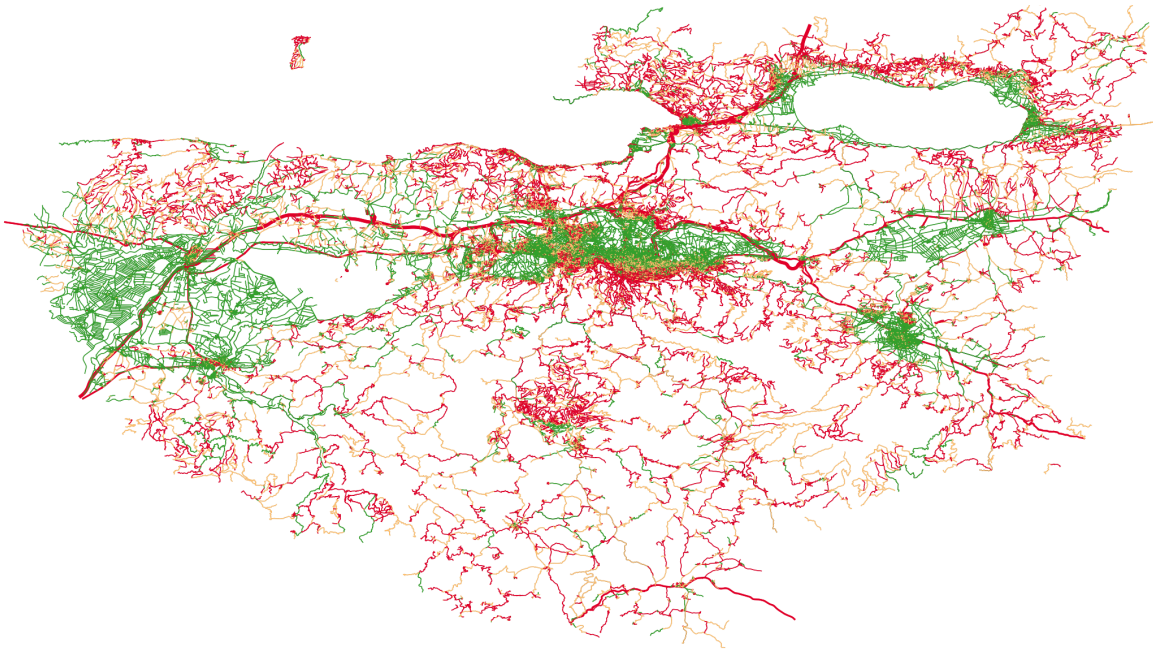


Figure 9: Bursa Whole City Biking Evaluation (With Highways and Trunk)